

Kashish Jaswani

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EDUCATION

Masters in Mathematics and Computing

2025 - 2027

Indian Institute of Science Education and Research, Pune, India

- *GPA (Current)* : 9.6/10
- *Courses in year 1*: Graph Theory, Algorithms, Rings and Modules, Fields and Galois Theory, Real Analysis II, Advanced Linear Algebra, Algebraic Number Theory, Complex Analysis, Measure and Integration.
- *Courses in year 2*: Probability, Ordinary Differential Equations, Statistical Inference.

Bachelors in Mathematics (Hons.)

2022 - 2025

Azim Premji University, Bangalore, India

- *GPA*: 8.85/10
- *Related coursework*: Major in Mathematics, with a minor in Media.
- *Courses (in Maths)*: Python Programming, Probability, Statistics, Mathematical Thinking, Real Analysis, Linear and Abstract Algebra, Multivariable Calculus, Differential Equations, Algorithms, Logic, Complex Analysis.
- *Honours Thesis Title*: Liar Games and Coding Theory

Schooling (Class 12)

2022

Delhi Public School, Pune, India

- *Percentage*: 99.4%
- *Board*: Central Board of Secondary Education
- *Subjects*: Applied Mathematics, Accountancy, Business Studies, Economics, English.

RESEARCH PROJECTS

Opinion Dynamics

- Working under the supervision of Prof. Moumanti Podder at IISER Pune.
- We are modelling an opinion dynamics graph and using probabilistic techniques to check if there is convergence of one of two opinions in the network (under assumptions that an existing opinion can change).

Inequality in academia from a network science perspective

- Done as summer research project under Prof. Barbara Keller at Aalto University, Espoo, Finland.
- We modelled the standard Barabasi Albert network with majority/minority partition and vertex removal. Then did inequality analysis on the resulting network.

Liar Games and Coding Theory

- Done as honours project under Prof. Ramanujam R at Azim Premji University.
- Looked at basics of Game Theory and Algebraic Coding Theory
- We looked at developing a one error correcting code based on the strategy to solve the liar game - where a lie by the respondent can be seen as an error in transmission.

Representation Theory and Tensor Products

- Done as summer research project by Indian Academy of Sciences under Dr. Keshab Chandra Bakshi at Indian Institute of Technology, Kanpur.
- Learnt basics of Representation Theory and Tensor Products.

PRESENTATIONS

- Poster Presentation on 'Liar Games and Coding Theory,' my honours project, given at Science Undergraduate Research Conclave conducted at Azim Premji University.
- Student talk on my honours project at the Student Research Conclave held at Azim Premji University in April 2025.

SKILLS

Computational

- Python
- SageMath

Typesetting

- $\text{\LaTeX}$

Languages

- English
- Hindi

WORKSHOPS

Theoretical Computer Science (TCS) Workshop

7 Dec 2024 - 10 Dec 2024

Indian Institute of Science, Bangalore, India

- Attended in depth lectures on foundational concepts of TCS.
- Attended lectures which focus on topics like Matching Theory and Differential Privacy.

Winter School

18 Dec 2023 - 27 Dec 2023

Punjab University, Patiala, India

- Attended the winter school by Punjab University and Indian Women and Mathematics.
- Special focus on Real Analysis, Linear Algebra, Combinatorics and Number Theory.

Summer School

May 2023 – June 2023

International Centre for Theoretical Sciences (ICTS), Bangalore, India

- Attended the Summer School for Women in Mathematics and Statistics conducted by ICTS.
- Special focus on Real Analysis, Linear Algebra, Statistics, R Programming (image processing).

EXTRACURRICULAR ACTIVITIES

- Peer tutor in Real Analysis for juniors.
- Part of the correction and invigilation team for the Regional Mathematics Olympiad.
- Participant in the Madhava Mathematics Competition 2025.
- Student editor of the second issue of the college zine, 'Math@apu.'
- Regular writer for the college newspaper, 'Namma Kathe' (both math and non-math articles).